

# Sarath Sivakumar

## AI Systems Engineer | Agentic AI & Backend Systems

Chennai, India | +91 9495683805 | sarathputhenvila@gmail.com  
linkedin.com/in/sarath-sivakumar | github.com/Sarath-Sivakumar

### PROFESSIONAL SUMMARY

---

Software engineer with 2+ years at Tata Consultancy Services building Java and Spring Boot backend services, with hands-on independent work in Agentic AI and LLM systems using Python and FastAPI. Architected an autonomous agent runtime covering planning, tool orchestration, memory and browser execution, with production-oriented runtime scalability, alongside RAG retrieval and AI developer tooling.

### TECHNICAL SKILLS

---

**AI Systems:** Generative AI, LLM applications, agentic AI, agent orchestration, planning, RAG, tool calling, function calling, structured outputs, context engineering, memory systems

**LLM Platforms & Frameworks:** OpenAI APIs, Anthropic Claude, Google Gemini, Ollama, LangChain

**Backend & Systems:** Python, FastAPI, Java, Spring Boot, microservices, REST APIs, WebSockets

**Data & Retrieval:** PostgreSQL, Redis, embeddings, semantic search, document chunking, reranking, pgvector, Chroma

**Automation:** Playwright, DOM analysis, element resolution, SPA and browser automation

**Cloud & DevOps:** Docker, Linux, Git, Jenkins, GitHub Actions, CI/CD

**Engineering Practices:** API design, distributed caching, performance optimization, load testing (JMeter), automated regression testing

**Languages:** Python, Java, SQL, JavaScript, TypeScript

### PROFESSIONAL EXPERIENCE

---

**Tata Consultancy Services** — Chennai, India

Apr 2024 – Present

*System Engineer*

Aug 2025 – Present

- Build and maintain 5+ Java and Spring Boot microservices supporting business-critical workflows; a platform-wide refactoring effort across 25+ modules improved module execution performance ~30%.
- Cut response times 20–25% on the highest-traffic services, verified with JMeter load testing, by profiling hot paths, tuning data access and eliminating redundant downstream calls.
- Automated build and deployment pipelines with Jenkins and Docker, cutting manual deployment effort ~40% and shortening release cycles.

*Software Engineer*

Apr 2024 – Aug 2025

- Delivered full-stack features (Spring Boot, React) on a platform with 10,000+ monthly active users; reduced latency 25–30% on the endpoints I owned through PostgreSQL query and service-layer optimization.
- Designed REST APIs load-tested to 2,000+ concurrent users and implemented a Redis distributed caching layer, improving high-traffic endpoint response times 30–40%.
- Modernized a legacy analytics platform used by 500+ users, cutting database queries 40–50% and remediating 10+ security vulnerabilities.

## SELECTED AI SYSTEMS PROJECTS (INDEPENDENT)

---

### **Autonomous AI Agent Runtime & Orchestration Platform** — independent R&D, in active development

*Python · FastAPI · LLM APIs · custom agent orchestration · Playwright · PostgreSQL · Docker*

- Architected a custom autonomous agent runtime and orchestration layer that interprets user goals, determines executable next actions, and routes capabilities through a provider-neutral tool layer with discovery and structured capability contracts.
- Engineered layered conversation and mission-context persistence with compression and retrieval, preserving relevant state across long-running executions and conversation continuation instead of losing it mid-run.
- Built recoverable browser execution with Playwright across DOM analysis, element resolution and SPA navigation, using observation-driven continuation to re-plan failed steps instead of terminating execution.
- Built an automated regression suite spanning planning, browser, search, filesystem and recovery paths to prevent previously resolved failure modes from reappearing.
- Redesigned runtime state and session continuity around shared external persistence, removing single-instance coupling in the WebSocket runtime and enabling horizontally scalable deployment behind a load balancer.

### **AI Developer Copilot — Spring Boot Root-Cause Analysis** — independent project

*FastAPI · Python · LLMs · RAG · React · TypeScript*

- Built a diagnostic pipeline — log ingestion, knowledge retrieval and failure-pattern classification, contextual LLM reasoning, structured root-cause output with remediation guidance — covering 100+ Spring Boot failure patterns.
- Grounded root-cause generation in a curated base of Spring Boot failure evidence, constraining diagnoses to known startup and dependency failure signatures, with analysis returned in under 3 seconds per log payload.
- Developed an interactive diagnostic interface surfacing root-cause analysis and fix steps, reducing manual log-investigation effort ~60% on the failure scenarios tested.

### **Conversational AI Agent POC — Multichannel Business Assistant** — independent proof of concept

*Python · LLMs · RAG · Telegram and WhatsApp APIs · Docker*

- Designed and implemented a proof-of-concept conversational assistant integrated with Telegram and WhatsApp, with LLM-driven multi-turn conversation handling and context tracking across sessions.
- Grounded responses in business-specific knowledge through a document ingestion and retrieval (RAG) pipeline; end-to-end responses remained below 5 seconds in tested interactions.
- Decoupled conversation orchestration from channel adapters so Telegram and WhatsApp share the same reasoning pipeline, allowing future transports such as telephony to integrate without redesigning conversation logic.

## EDUCATION

---

### **Indira Gandhi National Open University (IGNOU)**

Jul 2026 – Expected 2028

M.Sc., Artificial Intelligence and Machine Learning

### **St Berchmans Institute of Management Studies**

2019 – 2023

Bachelor of Computer Applications (BCA)

## CERTIFICATIONS

---

Microsoft Agentic AI Builder

GitHub Copilot Certified Developer

TCS Wings — Advanced Spring Boot